AWS NETWORKING PRACTICAL

S3 Gateway VPC Endpoint

Evidence-Based Implementation Report

Region: `ap-south-1` **VPC:** `10.0.0.0/16`

Evidence status: Configuration is documented; end-to-end transfers remain unverified.

1 Introduction

This practical documents an AWS design for storing an update file in Amazon S3 and making it available to an EC2 instance in a private subnet. Amazon S3 is object storage: a bucket holds objects, each addressed by its object key. The central networking component is an S3 Gateway VPC Endpoint. It routes supported S3 traffic from an associated VPC route table to S3 without sending that traffic through a public internet path.

This report separates the intended workflow from what the supplied console screenshots and configuration details actually establish. A resource appearing in a configuration screen or route table is not, by itself, proof that an EC2 transfer or connectivity test succeeded.

2 Objective

The objective is to configure a VPC with public and private subnets, give a public EC2 instance an IAM role for uploading an object to a private S3 bucket, and provide a private EC2 instance with a route to S3 through a Gateway VPC Endpoint. A temporary NAT Gateway was also considered for general outbound internet access, but the supplied NAT configuration contains a subnet discrepancy that prevents confirming working internet egress.

3 Prerequisites

Use an AWS account with permission to configure VPC networking, EC2, S3, IAM, VPC endpoints, Elastic IP addresses, and NAT Gateways. The operator should be familiar with IPv4 CIDR blocks, route tables, IAM permissions, and the distinction between public and private subnets. Access to an external update source is needed for the intended file workflow, but no source URL or download command is shown in the evidence.

4 Architecture Overview

The intended workflow is:

1. A public EC2 instance obtains a sample update file from an external source.
2. The instance uses temporary credentials from its attached `Endpoint-Uploader-Role` IAM role to upload the object to S3.
3. A private EC2 instance uses its own reader role to retrieve the object.

4. The private subnet's route table sends S3 traffic through the S3 Gateway VPC Endpoint. A public IP address and NAT Gateway are not required for this S3 path.

The Internet Gateway connects a VPC to the internet when routes and instance addressing allow it. A NAT Gateway can provide outbound internet access for instances without public IP addresses; it is not a general-purpose private link to AWS services. The S3 Gateway VPC Endpoint provides a private route to S3 only. In this configuration it is associated with the private route table, so it does not provide general internet access and does not establish an S3 endpoint route for the public subnet.

Access depends on more than the network route. The IAM identity policy on a role authorizes actions for that role; the S3 bucket policy is a resource-based policy that names allowed principals; and the VPC endpoint policy limits which requests may pass through that endpoint. These policy layers must all permit a request. An endpoint policy does not grant a role permission that its IAM policy or the bucket policy does not allow.

5 AWS Resources and Configuration

The table summarizes the reference configuration and identifiers visible in the supplied screenshots. The Internet Gateway name has a discrepancy, described in the table and later in the implementation.

Resource	Configuration and evidence
Region	Asia Pacific (Mumbai), `ap-south-1`.
VPC	`Endpoint-Lab-VPC`, ID `vpc-07be367d7afbe9118`, CIDR `10.0.0.0/16`.
Public subnet	`Endpoint-Public-Subnet`, ID `subnet-0e10f9294c2a40aad`, CIDR `10.0.1.0/24`. The EC2 configuration screen selects this subnet and enables automatic public IPv4 assignment.
Private subnet	`Endpoint-Private-Subnet`, ID `subnet-06c2277b86925498c`, CIDR `10.0.2.0/24`.
Internet Gateway	Reference configuration: `Endpoint-Lab-IGW`. Creation and listing screenshots show `Endpoint-IGW`, ID `igw-0d7170e3d7f2038a9`, attached to the VPC. The name mismatch cannot be resolved from the supplied evidence.
Public route table	`Endpoint-Public-RT`, ID `rtb-00a8f170bb7514b8d`, associated with the public subnet. Its routes include local VPC traffic and `0.0.0.0/0` to the Internet Gateway.
Private route table	`Endpoint-Private-RT`, ID `rtb-08163ee8c0a3b1eb2`, associated with the private subnet. Its S3 route is `pl-78a54011` to the Gateway Endpoint; a later screenshot also shows a default route to a NAT Gateway.
S3 Gateway Endpoint	`Endpoint-Lab-S3-Gateway`, ID `vpce-04e94ac9106c58510`, service `com.amazonaws.ap-south-1.s3`, type Gateway, state Available. Associated route table: `Endpoint-Private-RT`.
S3 bucket	`endpoint-lab-bucket-715615725174-ap-south-1-an`, in `ap-south-1`. Block all public access is enabled. The screenshots also show ACLs disabled and bucket versioning disabled.
S3 object	`updates/sample-update.txt`, approximately 4.6 KB, listed in the bucket console.
IAM roles	`Endpoint-Uploader-Role` and `EndpointLab-Reader-Role`. The EC2 configuration screenshot selects an instance profile named `Endpoint-Uploader-Role`.
Temporary NAT Gateway	Name `Endpoint-Temporary-NAT`, ID `nat-04c78399e2b8dd17f` in the later route screenshot. The creation screen selects public connectivity, zonal mode, an Elastic IP allocation, and `Endpoint-Private-Subnet`; that subnet selection is inconsistent with a working public NAT Gateway.

IAM permissions summary

The bucket ARN identifies the bucket itself. Object actions use an object ARN, which includes the bucket name followed by an object key or wildcard.

Policy layer	Action and resource	What the supplied configuration shows
Uploader role identity policy	`s3:PutObject` on `arn:aws:s3:::endpoint-lab-bucket-715615725174-ap-south-1-an/*`.	Authorizes the uploader role to write bucket objects, subject to the other policy layers.
Reader role identity policy	`s3:GetObject` on `arn:aws:s3:::endpoint-lab-bucket-715615725174-ap-south-1-an/*`; `s3:ListBucket` on `arn:aws:s3:::endpoint-lab-bucket-715615725174-ap-south-1-an`.	Object retrieval and bucket listing use different resource ARNs.
Bucket policy	Allows `s3:PutObject` to `Endpoint-Uploader-Role` and `s3:GetObject` to `EndpointLab-Reader-Role` on the bucket's object ARN pattern.	The visible policy screenshot contains these allow statements. It does not show a `aws:SourceVpce`-based explicit deny. It does not establish that reader access is limited exclusively to the VPC endpoint.
Endpoint policy	Allows `s3:ListBucket` on the bucket ARN, and `s3:GetObject` and `s3:PutObject` on its object ARN pattern.	The custom endpoint policy is an additional filter for requests that use this endpoint; it is not a substitute for role or bucket permissions.

The bucket policy screenshot identifies these role principals:

- `arn:aws:iam::715615725174:role/Endpoint-Uploader-Role`
- `arn:aws:iam::715615725174:role/EndpointLab-Reader-Role`

The endpoint policy uses a wildcard principal, but its action and resource list is limited to the specified bucket and its objects. A wildcard principal in an endpoint policy does not, on its own, make the bucket public or bypass IAM and bucket policies. Block Public Access is enabled.

6 Implementation Steps

6.1 VPC and subnet configuration

1. Create `Endpoint-Lab-VPC` in `ap-south-1` with IPv4 CIDR `10.0.0.0/16`. The creation screenshot selects a VPC-only setup and no IPv6 CIDR.

2. Create `Endpoint-Public-Subnet` with CIDR `10.0.1.0/24` and `Endpoint-Private-Subnet` with CIDR `10.0.2.0/24`. The public subnet is shown in Availability Zone `ap-south-1b` in its configuration screen.

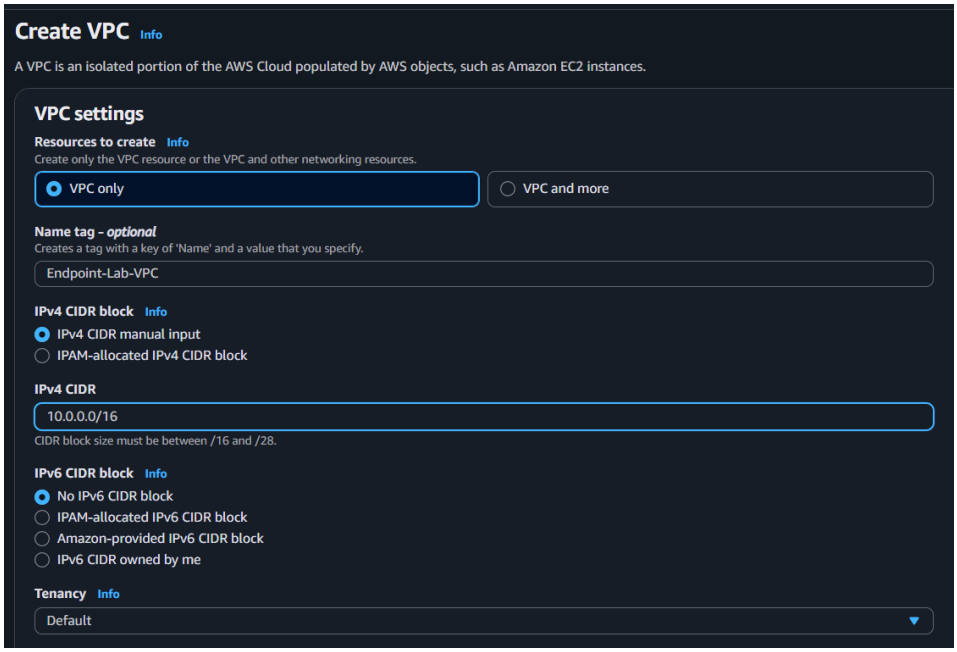


Figure 1: VPC creation settings for Endpoint-Lab-VPC and CIDR 10.0.0.0/16.



Figure 2: VPC summary showing Endpoint-Lab-VPC and its identifier.

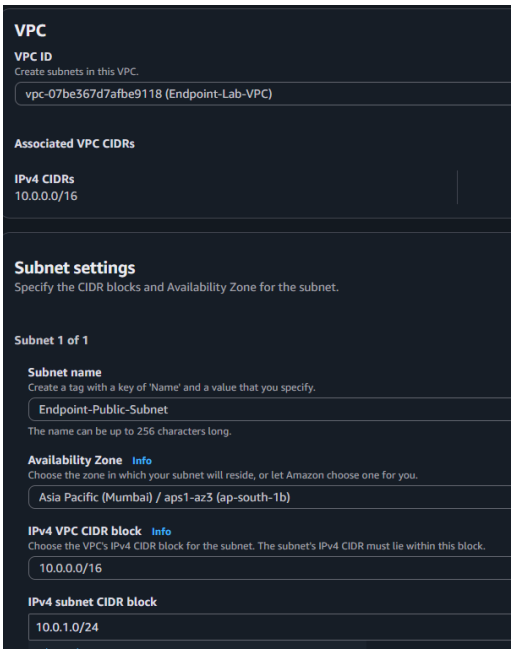


Figure 3: Public subnet configuration, including CIDR 10.0.1.0/24.

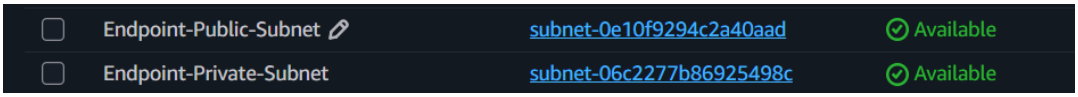


Figure 4: Subnet list showing the public and private subnet names and CIDR blocks.

6.2 Internet Gateway and route tables

3. Create and attach the Internet Gateway. The reference calls it `Endpoint-Lab-IGW`, while the supplied creation and listing screenshots show `Endpoint-IGW` (`igw-0d7170e3d7f2038a9`). This report preserves the visible screenshot name and records the mismatch rather than assuming the names are identical.

4. Create `Endpoint-Public-RT` and associate it with `Endpoint-Public-Subnet`. Its local route handles VPC traffic; its default route `0.0.0.0/0` targets the attached Internet Gateway. This makes the subnet public at the route-table level. Internet use by an EC2 instance also depends on public addressing and applicable security controls.

5. Create `Endpoint-Private-RT` and associate it with `Endpoint-Private-Subnet`. Initially, its shown routes are the VPC local route and the S3 prefix-list route to the Gateway Endpoint. A later route-table screenshot shows the added default route to the NAT Gateway as well. The private route table is visible in `endpoint/21.png` and its later route state in `endpoint/28.png`.

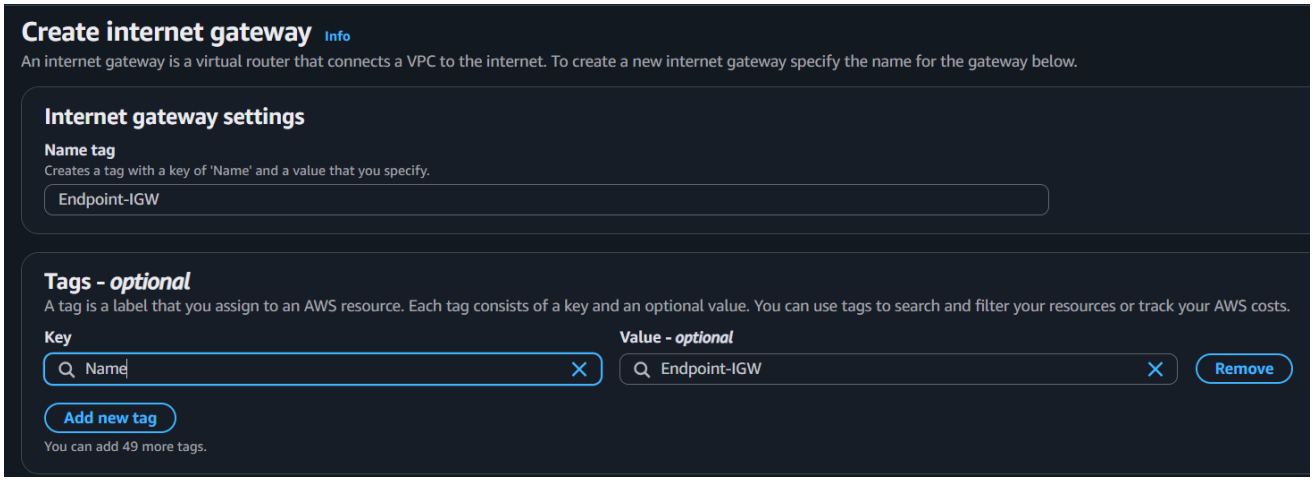


Figure 5: Internet Gateway creation screen showing the name Endpoint-IGW.



Figure 6: Internet Gateway listing showing Endpoint-IGW before attachment.

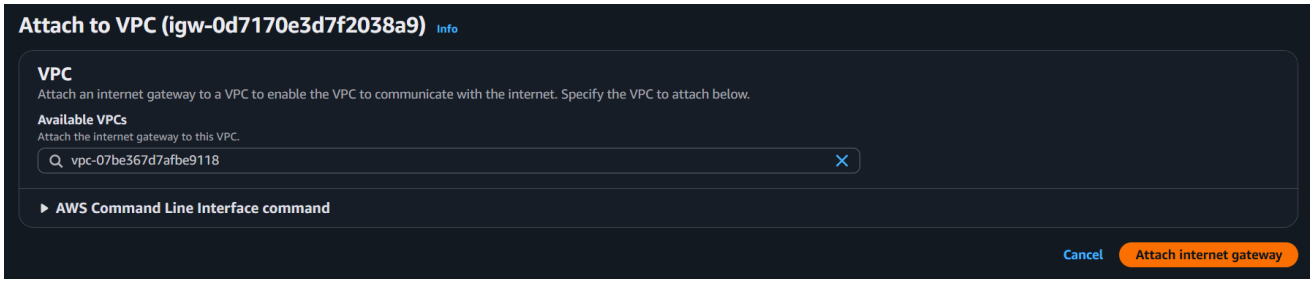


Figure 7: Internet Gateway attachment screen selecting the lab VPC.



Figure 8: Console status indicator showing the Internet Gateway as attached.

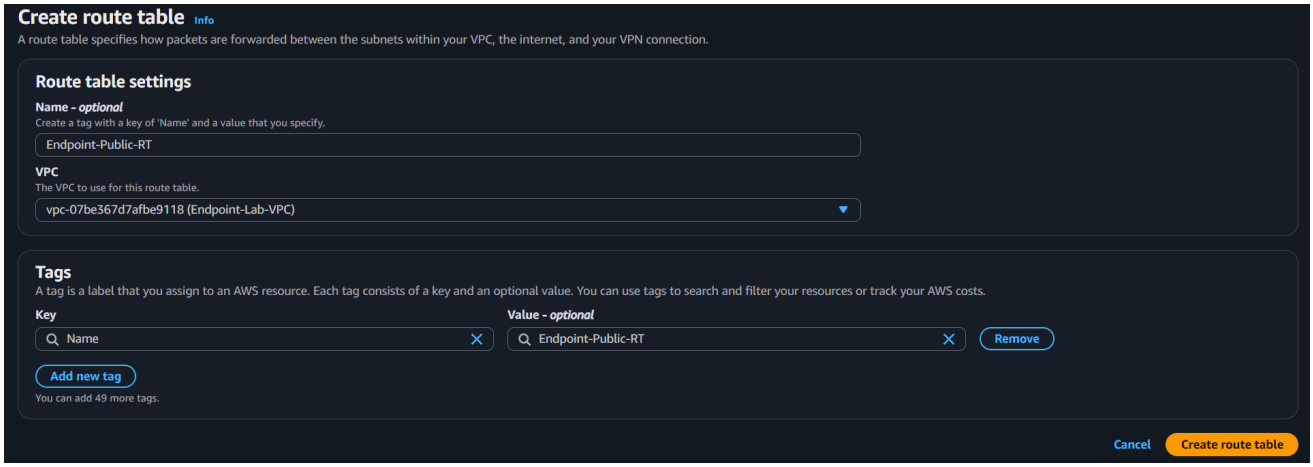


Figure 9: Creation settings for Endpoint-Public-RT.

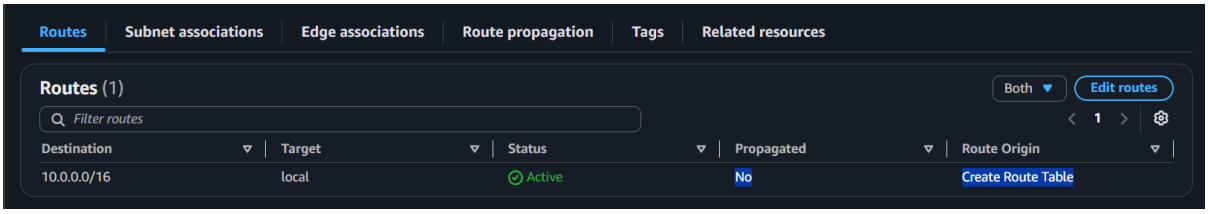


Figure 10: Initial route-table view showing the local VPC route.

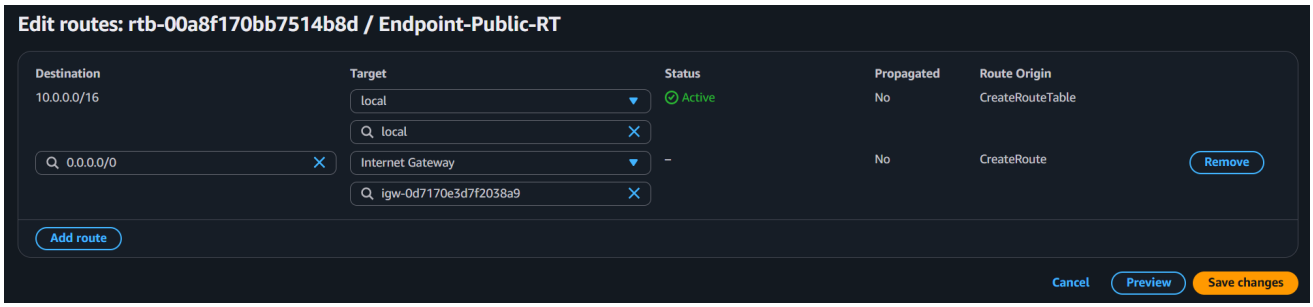


Figure 11: Public route table with the default route targeting the Internet Gateway.

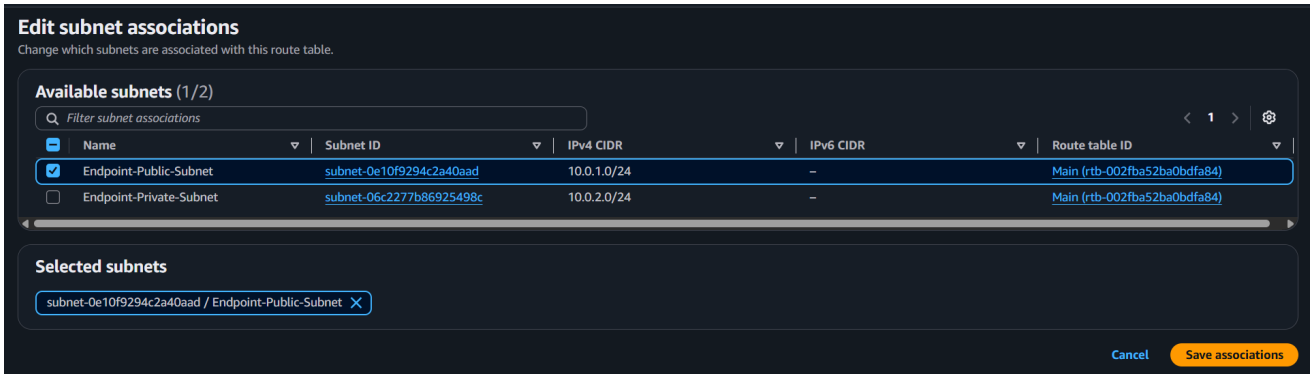


Figure 12: Subnet-association screen selecting Endpoint-Public-Subnet.

6.3 S3 bucket creation

6. Create the general-purpose S3 bucket listed in the resource inventory in `ap-south-1`. The bucket listing confirms its lowercase name; the create screen appears to show a differently capitalized prefix, so the discrepancy remains unresolved.

7. Keep Block Public Access enabled. ACLs are disabled with bucket-owner-enforced ownership, and versioning is disabled. The console lists the object at approximately 4.6 KB; see `endpoint/14.png`, `endpoint/15.png`, and `endpoint/26.png`.

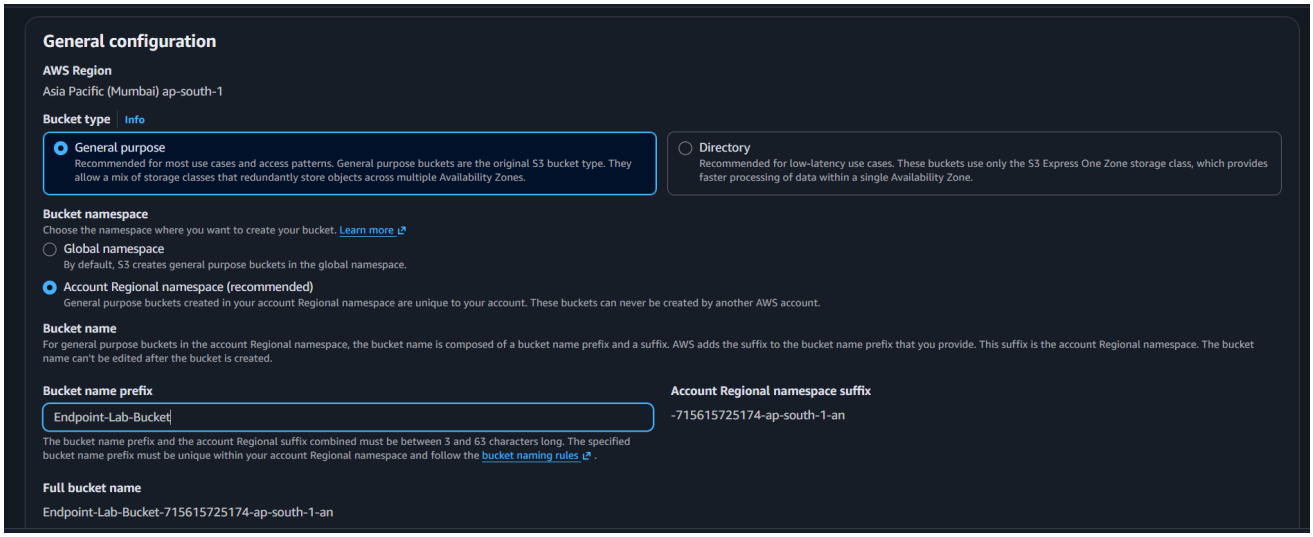


Figure 13: S3 bucket creation settings and the displayed bucket-name prefix.

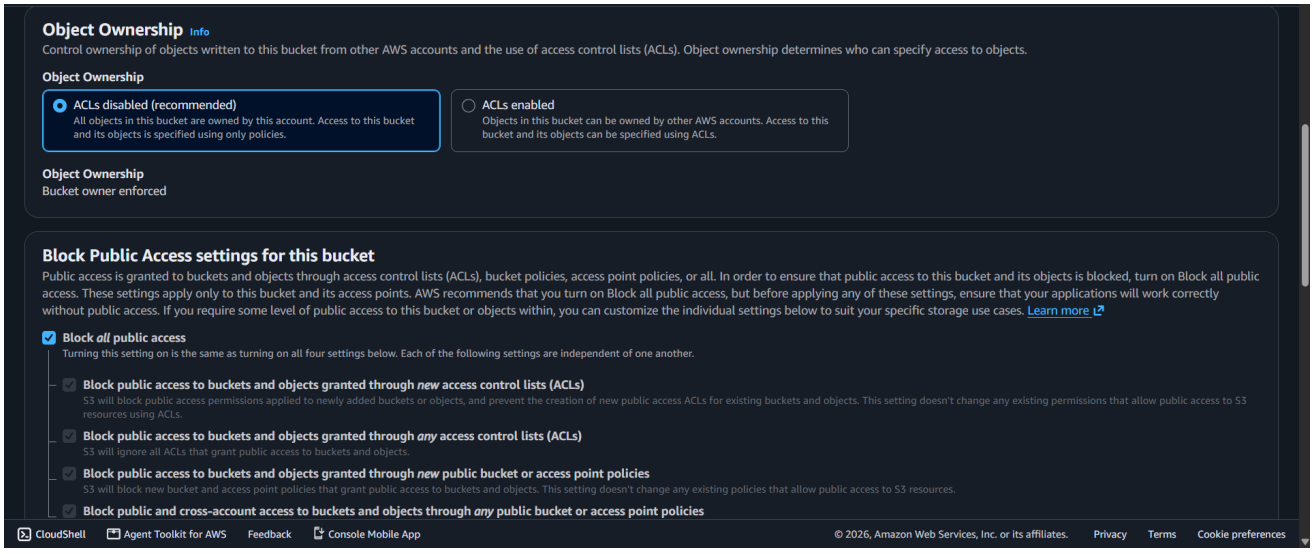


Figure 14: Bucket ownership settings and enabled Block Public Access controls.

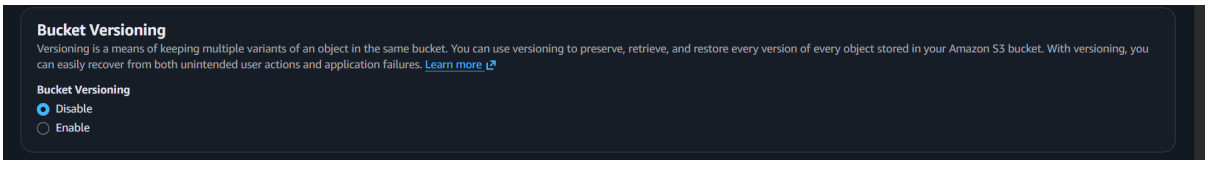


Figure 15: Bucket versioning setting shown as disabled.



Figure 16: S3 bucket listing showing the lowercase bucket name and Region.

6.4 IAM roles and permissions

8. Configure `Endpoint-Uploader-Role` with `s3:PutObject` access to the bucket's object ARN. The public EC2 setup screen selects an instance profile with the same name. An IAM role is the identity that carries permissions; an EC2 instance profile is the mechanism through which EC2 receives the role and its temporary credentials. The screenshot shows the selected profile, not the role's trust policy or proof that the instance was launched.

9. Configure `EndpointLab-Reader-Role` for `s3:GetObject` on the object ARN and `s3:ListBucket` on the bucket ARN. The supplied configuration shows a bucket-policy allow for this role to get objects. No screenshot establishes that the role was attached to a running private EC2 instance.

10. Apply a bucket policy that allows the uploader role to put objects and the reader role to get objects. The visible statements are resource-based allows. They do not include a visible explicit deny conditioned on the VPC endpoint.

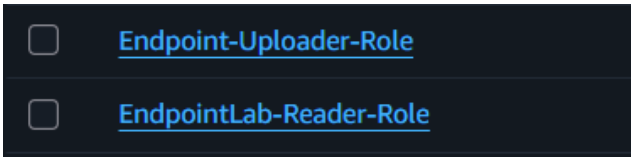


Figure 17: IAM role list showing Endpoint-Uploader-Role and EndpointLab-Reader-Role.



Figure 18: Bucket policy showing PutObject and GetObject allows for the named IAM roles.

6.5 S3 Gateway Endpoint creation

11. Create the S3 Gateway endpoint in `Endpoint-Lab-VPC` for service `com.amazonaws.ap-south-1.s3`. Its name is `Endpoint-Lab-S3-Gateway` and its ID is `vpce-04e94ac9106c58510`. Associate it with `Endpoint-Private-RT` and use the custom policy summarized above. The endpoint adds managed prefix-list route `pl-78a54011` to that route table. The reference configuration reports the endpoint as Available.

The prefix-list route is more specific than the default route. Therefore, where both routes are present, S3-bound traffic uses the Gateway Endpoint while other IPv4 destinations match the `0.0.0.0/0` route. This endpoint is specific to supported S3 traffic in the region; it is not an internet gateway or a general internet route.

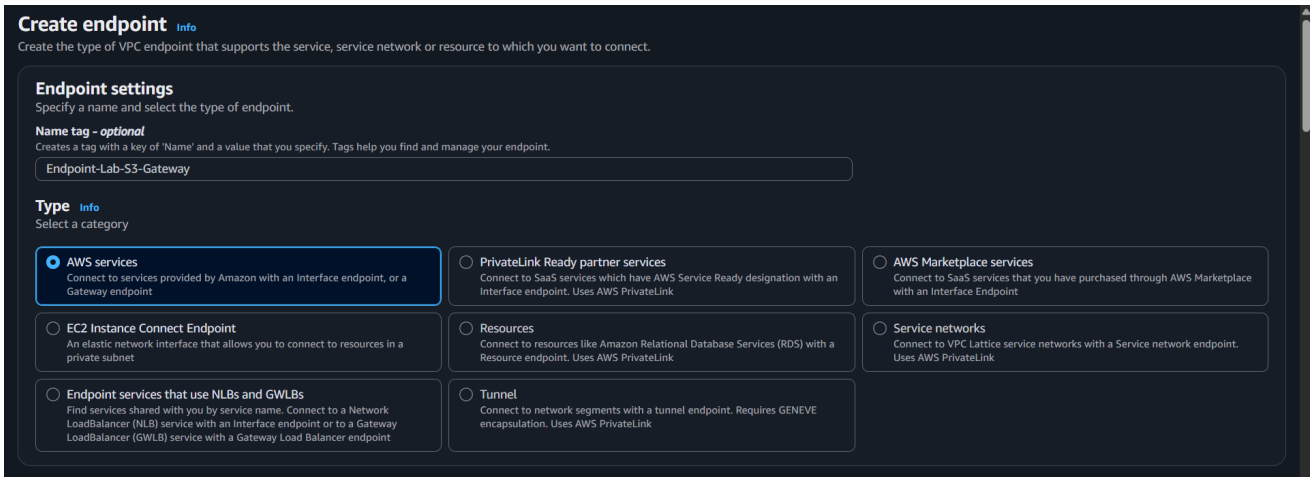


Figure 19: Gateway VPC Endpoint creation screen with the Endpoint-Lab-S3-Gateway name.

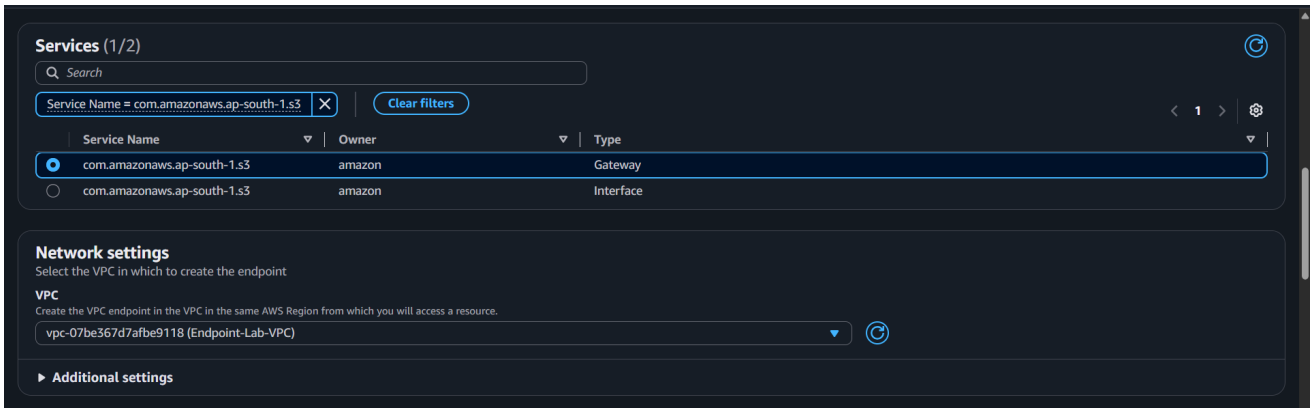


Figure 20: S3 service selection showing the Gateway endpoint type in ap-south-1.

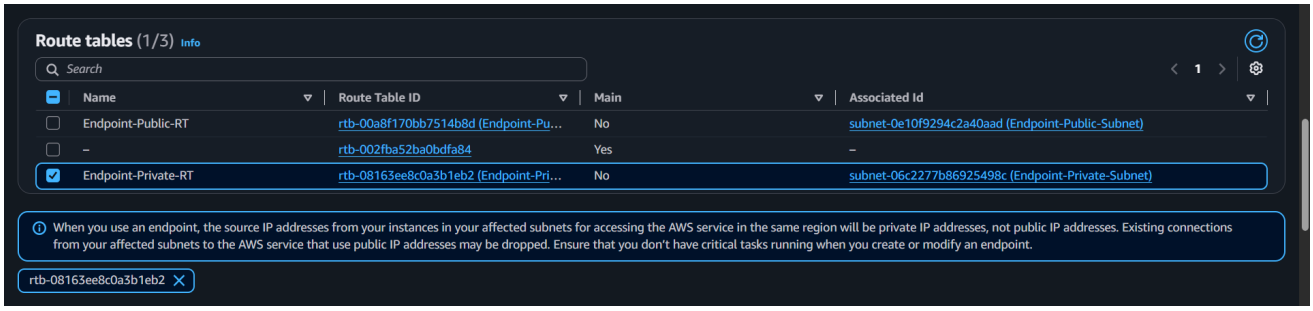


Figure 21: Route-table association showing Endpoint-Private-RT selected for the endpoint.

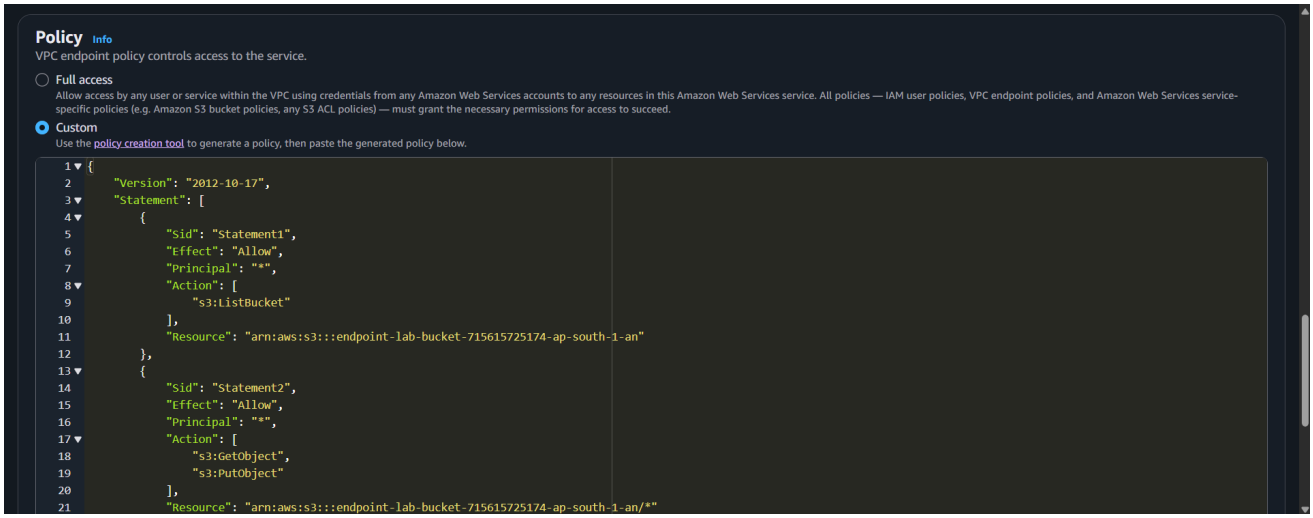


Figure 22: Custom VPC endpoint policy for bucket-level listing and object reads and writes.

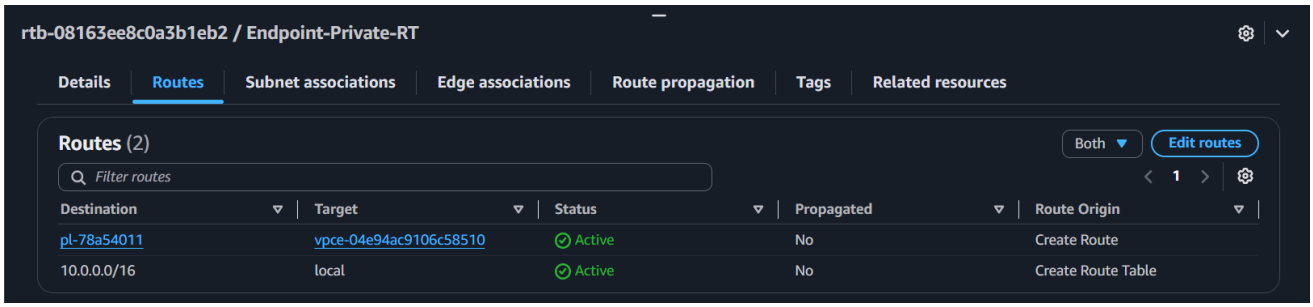


Figure 23: Endpoint-Private-RT showing the S3 prefix-list route to the VPC endpoint.

6.6 EC2 configuration

12. The public EC2 launch screen selects the VPC and public subnet shown in the resource inventory, enables automatic public IPv4 assignment, and selects key pair `my-k-key-ed25519` and the `Endpoint-Uploader-Role` instance profile. The screenshot shows launch settings, not a running instance or instance ID.

13. The intended private EC2 instance belongs in `Endpoint-Private-Subnet` and should use the reader role. A public IP is not needed to access S3 through the Gateway Endpoint. The supplied evidence does not show the private instance's launch configuration or confirm that it received the reader role.



Figure 24: EC2 advanced details with the Endpoint-Uploader-Role instance profile selected.

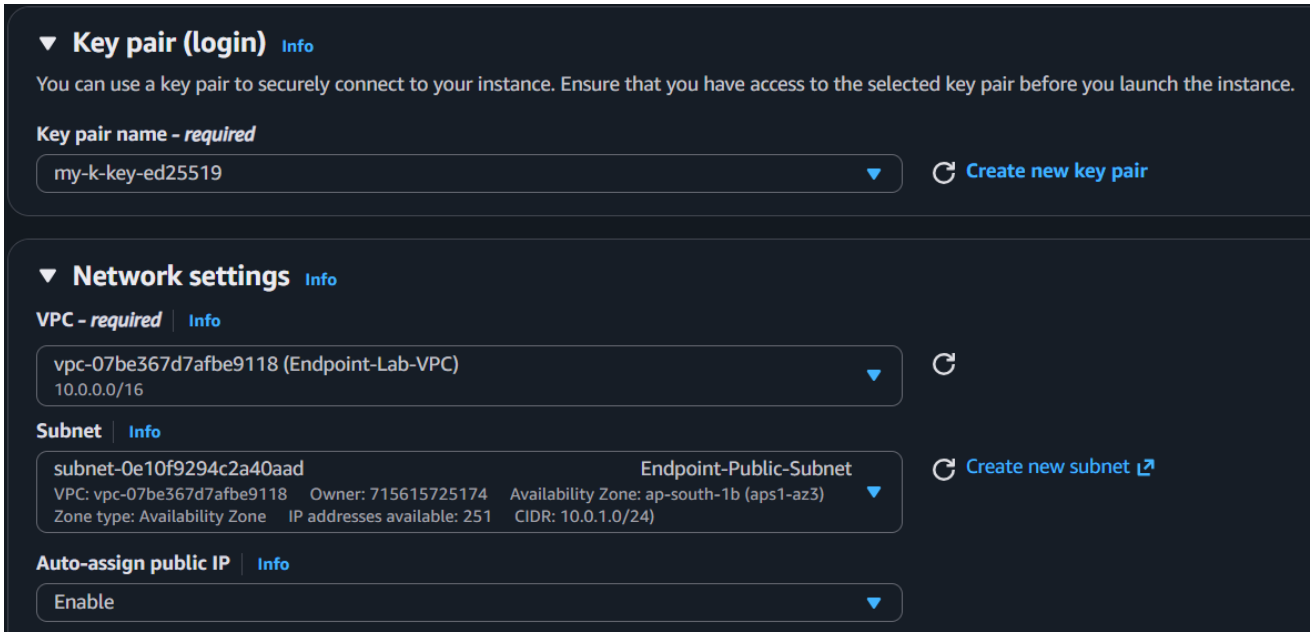


Figure 25: Public EC2 network configuration showing the VPC, public subnet, and public-IP setting.

6.7 File upload and retrieval workflow

The intended workflow is for the public instance to obtain an update file from an external source and write it to the bucket using its role, followed by retrieval from the private instance using its reader role. A private instance can reach S3 through the Gateway Endpoint without a public IP and without sending the S3 request through a NAT Gateway.

The evidence establishes only that an object named `updates/sample-update.txt` is present in the bucket listing. It does not show how the object arrived, a successful upload from the public EC2 instance, a download from the private EC2 instance, or the commands and output for either action. These transfers are therefore described as intended workflow, not as verified results.

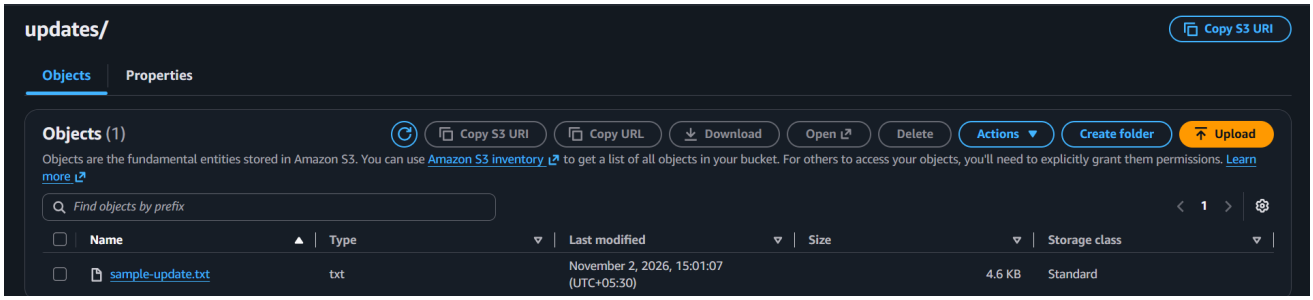


Figure 26: S3 object listing showing sample-update.txt in the updates/ prefix at about 4.6 KB.

6.8 Temporary NAT Gateway configuration and discrepancy

14. The NAT form shows `Endpoint-Temporary-NAT`, zonal mode, Public connectivity, Elastic IP allocation `eipalloc-065e91f5f78b4bcc9`, and subnet `Endpoint-Private-Subnet`. A public NAT Gateway belongs in a public subnet whose route table points to an Internet Gateway. The selected private subnet conflicts with that design. See `endpoint/27.png`.

15. The later private route table shows `0.0.0.0/0` targeting `nat-04c78399e2b8dd17f` with route status

Active. This establishes that the route exists and its target is shown as active in that screenshot. It does not prove the NAT Gateway was created in the correct subnet or that an instance obtained outbound internet connectivity.

For a working public NAT design, the NAT Gateway would be placed in `Endpoint-Public-Subnet`, with that subnet's route table leading to the Internet Gateway; the private route table would then send its default route to that NAT Gateway. This is the correct design principle, not a claim that the supplied NAT was configured that way. A correctly configured NAT Gateway may provide temporary general outbound access, for example for installing packages or the AWS CLI, but no such use is verified here. The S3 prefix-list route should remain in the private route table so S3 traffic continues to use the Gateway Endpoint. The Gateway Endpoint is sufficient for S3 and does not depend on the NAT Gateway.

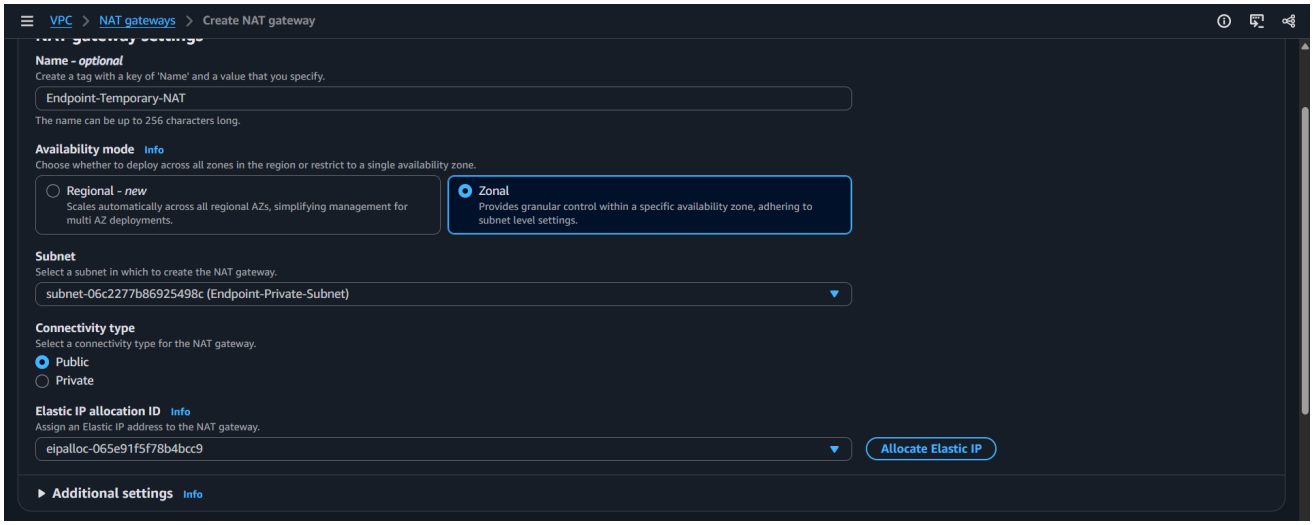


Figure 27: Temporary NAT Gateway configuration showing Public connectivity and the private-subnet selection.



Figure 28: Later private route-table view showing the S3 endpoint route, NAT default route, and local route as Active.

7 Verification and Observations

The supplied screenshots and reference configuration support these observations:

- The VPC and subnet configuration screens show `10.0.0.0/16`, `10.0.1.0/24`, and `10.0.2.0/24`; route-table associations show the public route table associated with the public subnet and the private route table with the private subnet.
- The public route table shows a default route to the Internet Gateway. The private route-table screenshot shows the S3 prefix-list route to `vpce-04e94ac9106c58510`; the later screenshot also shows the

NAT default route. Both routes are marked Active in that later view.

- The reference configuration identifies the S3 endpoint as Available. Route state and endpoint state are configuration observations, not application-level connectivity tests.
- Block Public Access is enabled, and the bucket console lists an object of approximately 4.6 KB.
- The EC2 launch screen shows public-subnet, public-IP, key-pair, and uploader-profile selections. It does not show a running instance or a completed transfer.

No supplied screenshot or output demonstrates a successful external download, S3 upload by the uploader instance, S3 retrieval by a private instance, or general outbound internet access through the NAT Gateway. No SourceVpce-based bucket-policy restriction is visible.

8 Limitations and Unverified Steps

- The reference names the Internet Gateway `Endpoint-Lab-IGW`; its creation and listing screenshots show `Endpoint-IGW`.
- The S3 bucket creation form appears to show a capitalized prefix, while the bucket listing and reference configuration show the lowercase bucket name used in this report. The evidence does not explain this difference.
- The NAT Gateway screen selects the private subnet even though the gateway is configured as Public. The later route being Active does not resolve the subnet discrepancy or prove egress works.
- The screenshots do not establish that either EC2 instance was launched and running, that the private instance had the reader role, or that IAM role trust relationships and security controls were tested.
- There is no transfer output or connectivity test for S3 or general internet access. The presence of the object in the S3 console alone does not identify the uploader or prove the intended EC2-to-EC2 workflow.
- The bucket policy screenshot does not show an explicit `aws:SourceVpce` deny. Reader access must not be described as restricted to the endpoint based on this evidence.

9 Conclusion

The configuration demonstrates the intended network design for private access to S3: an S3 Gateway VPC Endpoint is associated with the private route table, and the route table uses the S3 managed prefix list to direct S3 traffic to that endpoint. The bucket is not public, and the uploader and reader roles have distinct S3 permissions in the supplied configuration.

The evidence also reveals unresolved differences in the Internet Gateway name, the bucket creation form, and especially the NAT Gateway subnet selection. The later route-table entry and the visible S3 object confirm configuration and object presence, respectively, but they do not verify successful internet egress or the complete upload-and-retrieval workflow.

10 Glossary

EC2 instance
A virtual server running in an AWS VPC.

Elastic IP
A static public IPv4 address that can be associated with resources such as a public NAT Gateway.

Gateway VPC Endpoint
A VPC route-table target for supported AWS services, including S3. It provides service-specific routing, not general internet access.

IAM role
An AWS identity with permissions that trusted principals, such as EC2, can assume to obtain temporary credentials.

Instance profile
The EC2 attachment mechanism that carries an IAM role to an instance.

Internet Gateway
A VPC component that enables internet routing when attached to the VPC and selected by a route table.

NAT Gateway
A managed network address translation service that allows private-subnet resources to initiate eligible outbound connections without accepting unsolicited inbound internet connections.

Route table
A set of destination and target rules that determines where subnet traffic is sent.

Security group
A stateful virtual firewall applied to network interfaces; it controls allowed inbound and outbound traffic.

Subnet
A range of VPC IP addresses in one Availability Zone. A subnet is considered public when its route table has a route to an Internet Gateway; private subnets do not have that direct route.

VPC endpoint policy
A policy that controls which service actions and resources may be accessed through a VPC endpoint; it does not replace IAM or resource policies.

VPC
An isolated virtual network in an AWS Region.